1.2.4 Agentic AI and autonomous decision-making

Agentic AI represents a shift toward systems composed of software agents that autonomously reason, plan, and execute complex workflows across digital twins. These agents can monitor systems, learn from data, make decisions, and perform actions to achieve goals like reducing costs or optimizing operations, either individually or in collaboration with other agents.

Digital twins feed agentic AI the data it needs, while the AI enhances the twin through autonomous optimization and decision-making.

1.3 What makes a good digital twin?

When you hear "digital twin", you might picture a sleek 3D visualization of a complex machine. While visualizations are a common component, they are not the whole story. In fact, some digital twins operate as *headless twins*—systems focused purely on data processing, analytics, and automated decision-making without any visual component at all.

Digital twins range from simple monitoring dashboards to complex predictive and autonomous models. It is useful to categorize them by maturity to help identify the right goal for your needs and measure the expected business value. This book follows the five-level maturity model developed by Verdantix in a 2019 report titled "Smart Innovators: Digital Twins For Industrial Facilities", which categorizes digital twins from basic descriptive systems to advanced autonomous operations, as shown in figure 1.2.